I Remember That Question: Using Eye Tracking to See How Prequestions Shape Reading

Abstract

Answering questions before learning (prequestions) improves memory for the information they address, but how it does so is unclear. A common proposal is that prequestions direct attention to that information. We tested this account by recording eye movements while participants read a text about polio and tuberculosis after answering prequestions about one topic or none, so that every target sentence was read both with and without prequestions. Prequestions improved memory only for the prequestioned information, and eye movements showed the same specificity. Ex-Gaussian models, which separate typical reading time from occasional unusually long readings, showed that prequestions did not change typical reading of the answer sentences. Instead, these sentences received more unusually long readings, were looked back to more often, and were re-read for longer; other sentences were unaffected. Prequestions thus guide attention selectively rather than heightening it, consistent with additional processing when readers recognize an answer.

Keywords: prequestions, pretesting, eye tracking, reading, attention, learning

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A common study habit is to test oneself, retrieving answers to questions after a learning event (Hartwig & Dunlosky, 2012). Decades of research have shown how powerful this kind of retrieval practice is, a phenomenon known as the *testing effect*: retrieving studied information improves later memory more than restudying it does, across many types of materials and contexts (S. K. Carpenter, 2012; Dunlosky et al., 2013; Kornell & Vaughn, 2016; Roediger & Karpicke, 2006; Rowland, 2014). Answering questions *before* a learning event, however, can also enhance memory, even though learners usually do not know the answers and most of their guesses are wrong. Such *prequestions*, or pretests, reliably improve memory for the information they ask about (King-Shepard et al., 2025; St. Hilaire et al., 2024). That they help even though there is nothing yet to retrieve fits the broader finding that making learning more effortful, and even more error-prone, can benefit long-term retention (Bjork, 1994).

A typical prequestion experiment involves two groups: a group that receives prequestions and a control group that does not (S. K. Carpenter et al., 2023). The prequestion group first answers questions before a learning event and is encouraged to give its best guess even when the answers are unknown; the control group begins the learning event without answering any questions. After the learning event, both groups take a final test, which includes the questions that were prequestioned for the prequestion group and new, non-prequestioned questions. The typical finding is that the prequestion group remembers the prequestioned information better than the control group does—the *prequestion effect*. Because this benefit is specific to the information targeted by the prequestions, it is often called the *specific benefit* of prequestions (Pan & Carpenter, 2023). Prequestions sometimes also improve memory for information they did not address, a *general benefit*, but general benefits are typically smaller and appear to depend on the materials, such as short lecture videos (S. K. Carpenter & Toftness, 2017), on the kinds of questions used (Bostan & Ozcelik, 2026; St. Hilaire et al., 2019), and on where untested

information appears relative to the tested information (Sana & Carpenter, 2023). Two recent meta-analyses confirmed a robust specific benefit ($g = 0.54$, St. Hilaire et al., 2024; $g = 0.66$, King-Shepard et al., 2025) but found no reliable general benefit; in the larger, multilevel analysis, the general benefit was essentially zero ($g = 0.01$, King-Shepard et al., 2025). The specific benefit, in contrast, has been replicated with prose passages (Richland et al., 2009) and video-recorded lectures (S. K. Carpenter & Toftness, 2017; St. Hilaire & Carpenter, 2020), and has been examined in classroom settings (S. K. Carpenter et al., 2018; Masked Citation, n.d.; Soderstrom & Bjork, 2023; see Pan & Carpenter, 2023 for a review).

How exactly do prequestions benefit learning? Several non-mutually exclusive mechanisms have been proposed (King-Shepard et al., 2025; Pan & Carpenter, 2023): prequestions may stimulate curiosity about the answers (Berlyne, 1954, 1962), prompt elaboration by activating related knowledge (Kornell et al., 2009), or provide a metacognitive "reality check" that shows learners what they do not know (Bjork et al., 2013). Each of these would lead learners to allocate more attention to the information that answers the prequestions (King-Shepard et al., 2025). On this *attention-based* account, prequestions signal what is likely to be important on an upcoming test, so learners notice that information when it appears, process it more, and remember it better as a result (Pan & Carpenter, 2023). Eye-tracking studies largely agree: readers fixate prequestioned information for longer (Bostan & Ozcelik, 2026; Yang et al., 2021) and spend more time re-inspecting it and looking back to it (Lewis & Mensink, 2012).

A simple attention-based account, however, cannot fully explain the effects of prequestions. First, if prequestions only signalled what to attend to, telling learners what will be tested should work as well as asking them. Yet answering prequestions improved memory more than studying the material with the tested concepts emphasized or reading the questions without answering them (Richland et al., 2009), and more than reading learning objectives or statements of the target facts (Sana et al., 2020; but see Rivers et al., 2026). Second, prequestions reliably

improve memory for the prequestioned information (King-Shepard et al., 2025; St. Hilaire et al., 2024), but some studies also find a general benefit for information that was not prequestioned (Bostan & Ozcelik, 2026; Rivers et al., 2026; Sana & Carpenter, 2023; St. Hilaire et al., 2019). An account of prequestions must therefore explain why the benefit is sometimes specific and sometimes general. One such account is the attentional-window account (Sana & Carpenter, 2023), on which answering prequestions opens a window of heightened attention that closes once the prequestioned information has been found. Everything read while the window is open benefits, so the placement of the prequestioned information should matter. Sana and Carpenter (2023) tested this by manipulating where the prequestioned information appeared in a text. When it came at the beginning, prequestions produced only a specific benefit; when it came at the end, so that learners were still searching for the answers while reading the rest of the text, prequestions also produced a general benefit. Most of the evidence for these accounts, however, comes from what learners remember afterwards, not from how they process the material as they learn.

Eye tracking offers a direct way to test this. Whether reading a text or watching a film or video, the eyes alternate between brief pauses, called fixations, and rapid movements between them, called saccades. Where and for how long people fixate is thought to reflect the overt allocation of attention and the cognitive processes that support comprehension and learning (Deng & Gao, 2023; Hyönä et al., 2003; Madsen et al., 2021; Rayner, 1998; Rayner, 2009). Because eye tracking records this behavior continuously and without interrupting the reading or viewing experience, processing can be examined as it unfolds rather than inferred afterward from a memory test (Rayner, 1998). Eye movements can also distinguish processing when information is first encountered from processing when people later return to it, such as rereading a passage or revisiting a region of the screen. Eye-movement behavior is also sensitive to individual differences, including reading ability (Ashby et al., 2005; Perfetti, 2007), prior knowledge

(McNamara & Magliano, 2009), and working memory capacity (Daneman & Carpenter, 1980). For these reasons, real-time measures such as eye tracking have been identified as a key direction for prequestion research (Pan & Carpenter, 2023). In their meta-analysis, King-Shepard et al. (2025) specifically recommended using eye tracking to observe where learners spend more or less time on passages containing prequestioned and non-prequestioned information, which is the approach taken here.

Few studies have used eye tracking to examine prequestions during reading. In an early study, readers given questions before reading re-read and processed the question-related material more carefully and read more quickly through unrelated material (Rothkopf & Billington, 1979). Lewis and Mensink (2012) examined prequestions in two eye-tracking experiments with texts on different topics. In Experiment 1, students read a text about space travel after reading one of two sets of six prequestions, or no prequestions. Readers given one set spent longer re-inspecting the sentences relevant to those questions and looking back to them, and recalled more of them, than readers given the other set; the other set of questions had no reliable effect. In Experiment 2, whose text we used in the present study, students read a text about the humanitarian efforts of the Gates Foundation, which also described polio and tuberculosis (TB). One group read four prequestions about TB before reading (e.g., "How is tuberculosis spread through the air?"), and the other group read the passage without prequestions, while eye movements were recorded with a head-mounted eye tracker. Prequestions did not slow forward reading of the targeted sentences (e.g., "Tuberculosis is spread through the air when people who have the disease cough, sneeze, or spit"), but readers who received prequestions re-read them relatively more during first-pass reading, before moving on, and spent more time looking back to them after moving on to later parts of the passage. They also looked back somewhat more to the rest of the text. On a free-recall test, the prequestion group recalled about twice as much of the targeted information as the control group, but no more of the other information. The specific benefit thus occurred without a

general benefit, and it was accompanied by a shift of attention toward prequestioned information. Notably, this shift appeared only after the targeted sentences had been read once: readers did not slow down on first encountering them but went back to them, as if they first recognized the sentences as relevant and then returned to process them further.

More recently, Bostan and Ozcelik (2026) gave 12 adults *integrative* prequestions—questions that require combining information from different parts of a text—before they read a passage about car brakes, and compared them with 12 adults who read without prequestions. The prequestion group made more fixations and spent more time on prequestioned sentences, and made more transitions between them, than the control group. Unlike most prequestion studies, they also found a general benefit: the prequestion group scored higher on non-prequestioned questions, even though they spent less time on non-prequestioned sentences. These findings are consistent with an attention-based account.

Several studies have shown that attention to the prequestioned information increases following prequestions (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012; Yang et al., 2021). Whether this increase causes the benefit of prequestions or follows from it, however, remains unclear, and several findings show that prequestions can improve learning without a corresponding increase in attention to the material. In a large multi-site classroom study, prequestions before online lecture videos improved learning even though students watched less of the videos afterwards, and some students skipped the videos entirely after seeing prequestions (Motz et al., 2026). In the eye-tracking studies reviewed above, prequestions did not slow forward reading of the prequestioned sentences; instead, readers re-read them and looked back to them (Lewis & Mensink, 2012), and integrative prequestions improved memory for non-prequestioned information even though readers spent less time on it (Bostan & Ozcelik, 2026). To account for findings like these, Motz et al. (2026) proposed the *mental model hypothesis*: prequestions activate relevant knowledge before learning, which then facilitates the processing

and encoding of related information. When learners encounter content related to the prequestions, it resonates with the activated knowledge, and once they recognize it as related, they may direct additional attention to it. On this view, attention follows recognition of related content rather than directly causing the benefit. Consistent with this, the benefit of prequestions in Motz et al.'s study was larger for students who answered more of the prequestions correctly, suggesting that learners benefit most when they have relevant knowledge to activate.

The Current Study

The current study tested the attention-based account with a design that addresses several limitations of the two previous eye-tracking studies of prequestions during reading (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012). Participants read the Gates Foundation text used by Lewis and Mensink (2012), which discusses efforts to eradicate two diseases, polio and tuberculosis (TB). First, in Experiment 2 of Lewis and Mensink (2012) and in Bostan and Ozcelik (2026), one fixed set of sentences was prequestioned, so the effect of prequestions, and the absence of a general effect, could not be separated from the content of those particular sentences. Experiment 1 of Lewis and Mensink (2012) used two sets of prequestions that targeted different sentences, but the sets targeted different numbers of sentences, and only one of them affected reading. Here, the prequestioned content was fully counterbalanced: two prequestion groups were prequestioned on different topics, so that every target sentence was read both with and without prequestions. Second, the previous studies compared small groups—about 20 readers per group in Lewis and Mensink (2012), and 12 per group with two prequestions in Bostan and Ozcelik (2026)—so their estimates are imprecise. In addition, Bostan and Ozcelik (2026) compared readers' total fixation times and counts, so differences in how the groups read the same sentences could not be separated from differences between the readers themselves. Here, the sample was substantially larger—138 participants, about 46 per condition—and the models included variation across sentences and test questions as well as across readers, so that

the effects generalize beyond these particular items. Third, the previous studies analysed average reading or fixation times; we also modelled the distribution of reading times, which distinguishes slower reading overall from occasional unusually long readings. Fourth, in Lewis and Mensink (2012), readers were given the questions and asked to read the text to answer them, but did not answer them before reading. Readers may have answered them covertly, which can be as effective as answering prequestions overtly (Rivers et al., 2026), but whether they did is unknown. Here, as in most prequestion studies, participants answered each prequestion before reading. Finally, unlike the integrative prequestions of Bostan and Ozcelik (2026), each prequestion was answered by a single sentence of the text, as in Lewis and Mensink (2012), the kind of prequestion that typically produces a specific but not a general benefit (St. Hilaire et al., 2019). Before reading, participants answered five prequestions about polio, five prequestions about TB, or no prequestions (no-prequestion condition). Each of ten target sentences in the text contained the answer to one question. Because the Polio and TB conditions swapped which topic was prequestioned, every target sentence served as prequestioned information for one group of readers and as non-prequestioned information for another. Pooling the two prequestion conditions, reading of prequestioned sentences could therefore be compared with reading of the very same sentences without prequestions (the no-prequestion condition), with each topic contributing equally.

We examined two kinds of eye-movement measures. *Global* measures—total gaze duration, re-reading time, and regressions-in (look-backs)—reflect all processing a sentence receives, including later returns to it. The *local* measure—selective regression-path duration, our measure of first-pass reading—reflects processing when a sentence is first encountered, before the reader moves on to later text. Distinguishing the two indicates *when* prequestions change processing: as readers first encounter the answer, or when they return to it later.

Fixation and reading times are not normally distributed: they cannot fall below zero and are strongly right-skewed, with a long tail of unusually long readings (Balota & Yap, 2011; Staub et al., 2010). Analyses that assume normality therefore misdescribe them, and transforming the data to reduce the skew changes the hypotheses being tested (Lo & Andrews, 2015). We therefore analysed total gaze duration with the ex-Gaussian distribution, which describes durations as the sum of a Gaussian component, with mean μ, and an exponential component, with mean τ, and so models the skew directly. Beyond fitting the shape of the data, the ex-Gaussian has a second advantage: it separates two ways in which reading times can change, a shift of the whole distribution (μ) and a lengthening of its tail (τ), which an analysis of means cannot tell apart. Although the ex-Gaussian is a descriptive model of response times, its parameters have often been mapped onto cognitive processes, though these mappings remain debated (Matzke & Wagenmakers, 2009). The Gaussian component (μ) is typically linked to relatively automatic, early-stage processing, whereas the exponential component (τ) is linked to later, more controlled processes such as attentional control, working memory updating, error monitoring, and strategic processing. In eye-tracking research on reading, word frequency affects τ as well as μ of fixation durations (Staub et al., 2010), whereas a modest degradation of the text, such as presenting it faintly, shifts only μ (White & Staub, 2012); in scene viewing, the type of image shifts μ, whereas familiarity with the image lengthens only a subset of relatively long fixations, increasing τ (Guy et al., 2020). Some manipulations, then, lengthen a subset of unusually long fixations rather than shifting the whole distribution. Previous eye-tracking studies of prequestions analysed average reading or fixation times (Bostan & Ozcelik, 2026; Lewis & Mensink, 2012; Yang et al., 2021). Averages can mask effects of this kind: if prequestions add processing to only some encounters with a sentence, the mean changes little, and a longer tail cannot be distinguished from slower reading overall (Guy et al., 2020). Here, sentences that readers recognize as answering a prequestion may trigger reanalysis or additional processing. If

so, prequestion effects should appear primarily in τ rather than μ. Because μ and τ are both parameters of the ex-Gaussian model, we examined both: an effect on μ would indicate that prequestions shift typical reading of the prequestioned sentences as a whole, whereas an effect on τ would indicate that they lengthen only a subset of readings, the unusually long ones, consistent with additional processing on some encounters.

Hypotheses

For final test performance, we compared memory for prequestioned and for non-prequestioned material with memory for the same material in the no-prequestion condition, overall and on each topic. We predicted a specific benefit—better performance than the no-prequestion condition on the prequestioned topic—but no general benefit on the non-prequestioned topic.

For eye movements, we predicted a specific effect: prequestioned information would receive longer gaze durations and longer re-reading than non-prequestioned information and than the same information read without prequestions (the no-prequestion condition). Broken down by topic, readers given Polio prequestions should spend longer on Polio than on TB sentences, readers given TB prequestions should show the reverse pattern, and no-prequestion readers should show no difference between the two topics.

Method

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. The study was preregistered (https://osf.io/8env4/overview?view_only=5073f3b1114944c1a6c63339132a5ad5). All data, materials, and analysis scripts and code are available at https://anonymous.4open.science/r/Prequestions_EyeTracking-B118/; every number, table, and figure in this article is computed from the raw data by the code in that repository. The author used generative AI (Claude Opus 5.5, Anthropic) to draft and edit text and to check references; the author verified all content and citations and takes full responsibility for the article.

Participants

The target sample size was based on the specific benefit of prequestions. A recent meta-analysis reported a medium-sized specific benefit ($g = 0.54$; St. Hilaire et al. (2024)), corresponding to an odds ratio of 2.66. A simulation-based power analysis with mixedpower (Kumle et al., 2021) for the preregistered binomial mixed model, based on pilot data, indicated that 45 participants per group would provide 90% power to detect an effect of this size. We therefore planned to test 45 participants in each of the three conditions (135 in total).

Participants were recruited from the university SONA subject pool and received course credit. Eye-tracking data were recorded from 161 participants. One participant had eye-tracking data for fewer than all four pages of the text and was excluded. Eighteen participants did not meet the preregistered calibration criterion (accuracy and precision of 0.5° or better) or could not be calibrated and were excluded. Four participants had no final test data and were excluded. The final sample comprised 138 participants (no prequestions: 47; Polio prequestions: 45; TB prequestions: 46). Calibration validation data were available for all included participants. Mean accuracy was 0.37° ($SD = 0.10$, range 0.13–0.50), and mean precision (SD of gaze samples) was 0.22° ($SD = 0.12$, range 0.06–0.49). All included participants with calibration data met the criterion. The preregistered exclusion of participants with high prior knowledge was not applied.

Materials

Participants read a 690-word, multi-topic text (Lewis & Mensink, 2012) that discussed the Gates Foundation's humanitarian efforts in eradicating Polio and TB in the developing world. The text consisted of 38 sentences presented over four pages, each 16–17 lines long (Figure 1). Ten sentences served as target sentences: five about Polio ($M = 19.4$ words, $SD = 11.4$, range 4–35) and five about TB ($M = 16.2$ words, $SD = 3.7$, range 11–21). The Polio target sentences had a higher reading level than the TB target sentences, as measured by the Flesch–Kincaid grade level (Kincaid et al., 1975), computed with quanteda (Benoit et al., 2018): Polio $M = 14.1$, $SD = 6.2$; TB $M = 8.8$, $SD = 2.0$. Their syntactic complexity, estimated from a dependency parse with

UDPipe (Straka & Straková, 2017; Wijffels & Straka, 2026) and its model of the English Web Treebank (Silveira et al., 2014), was similar: mean dependency distance (Liu, 2008), the average number of words between each word and its syntactic head (Polio $M = 2.72$, $SD = 1.03$; TB $M = 2.62$, $SD = 0.40$), and number of clauses per sentence (Polio $M = 1.6$, $SD = 0.9$; TB $M = 1.8$, $SD = 0.8$). Each target sentence contained the answer to one of ten short-answer questions (five Polio, five TB; Appendix A). Questions were percentage-based, yes/no, or multi-part, and were worth 12 points per topic (24 points in total).

There were three conditions. In the Polio prequestion condition, participants answered the five Polio questions before reading; in the TB prequestion condition, they answered the five TB questions; in the no-prequestion condition, they read the text without prequestions. For each prequestion condition, its five target sentences were *prequestioned* and the other topic's five were *non-prequestioned*, so every target sentence served in both roles across conditions.

Apparatus

Eye movements were recorded with a Tobii Pro Spectrum eye tracker sampling at 600 Hz. The text was displayed on an EIZO FlexScan EV2451 monitor (23.8-inch, 1920 × 1080 pixels) viewed from approximately 65 cm, without a chin rest. The text was displayed in 36-point Courier New, a monospaced font, on a grey background. At this distance, one character subtended approximately 0.5° of visual angle and one line spacing approximately 1.1°.

The whole experiment was run in Tobii Pro Lab (version 25.19). The prequestions, distractor task, and final test were PsychoPy tasks hosted on Pavlovia and presented as web elements within the Tobii Pro Lab timeline.

Procedure

At the start of the session, each participant was randomly assigned to one of the three conditions (Polio prequestions, TB prequestions, or no prequestions). The session had four phases—prequestions, reading, a distractor task, and the final test—and participants in the no-prequestion condition skipped the first.

Prequestions

Participants in the two prequestion conditions answered the five short-answer questions on their topic (Polio or TB), in a random order ([masked for review]). They were told that they might not know the correct answers but should give their best guess. Each question was shown on the screen with a box below it, into which participants typed their response. They had unlimited time and pressed a triangle at the bottom of the screen when they were finished. No feedback was given.

Reading

Before reading, the eye tracker was calibrated with Tobii Pro Lab's 9-point calibration procedure (white targets on a grey background), and calibration accuracy and precision were validated and recorded. Participants then read the text, presented over four pages, while their eye movements were recorded.

Distractor Task

After reading, all participants completed the president recognition task ([masked for review]), a distractor task lasting approximately 3 minutes.

Final Test

Participants then answered all ten final-test questions—five on Polio and five on TB—in a random order ([masked for review]), in the same format as the prequestions, with unlimited time and no feedback. For participants in the prequestion conditions, five of the questions were the prequestions they had answered before reading, and five were new. Finally, participants were thanked and debriefed.

Scoring

Prequestion and final test responses were scored by Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`). Rather than judging responses one at a time, Claude wrote a deterministic, rule-based scoring script (`scoring_rubric.R`, with `grade_prequestions.R` and `grade_test.R`) that awarded one point for each element of the correct answer present in a

response, and applied it identically to every response. The scoring rules for each question are given in Appendix A. To check the automated scoring, a research assistant who was blind to condition, test phase, and the automated scores independently scored all 1,678 responses that were not blank. The two scorings agreed exactly on 97.6% of responses, and agreement was excellent, ICC(2,1) = .99, 95% CI [.99, .99], using a two-way random-effects, absolute-agreement, single-rater ICC (Koo & Li, 2016; Shrout & Fleiss, 1979), computed with the irr package (Gamer et al., 2026). For participants' final test totals, ICC(2,1) = .99. Agreement for each question is given in Table A2. All analyses used the automated scores.

Eye-Movement Data Processing

Sentence-level measures were computed from the fixation-level data rather than taken from Tobii Pro Lab's sentence metrics, for two reasons. First, when a page appeared, readers' eyes typically landed partway down the page before moving to the first line, (on 79% of pages, the first fixation on the text fell below the first sentence), so Tobii Pro Lab treated the sentences above the landing point as already passed and recorded no first-pass reading for them. Second, recorded gaze positions were often offset vertically by up to about half a line, so fixations were sometimes assigned to the sentence on a neighbouring line.

Fixations were identified with Tobii Pro Lab's Tobii I-VT (Fixation) filter, a velocity-threshold identification algorithm (Olsen, 2012; Salvucci & Goldberg, 2000), with the following settings. Gaps in the gaze data shorter than 75 ms (e.g., brief losses of tracking) were filled by linear interpolation, the gaze positions of the two eyes were averaged, and noise was reduced with a moving median over a window of three samples. A gaze sample was classified as part of a fixation when the angular velocity of the eye, computed over a 20-ms window, was below 30°/s. Adjacent fixations separated by less than 75 ms and less than 0.5° of visual angle were merged, and fixations shorter than 60 ms were discarded. Fixations cut off when a page appeared or disappeared can nonetheless be shorter than 60 ms in the export. We excluded fixations shorter

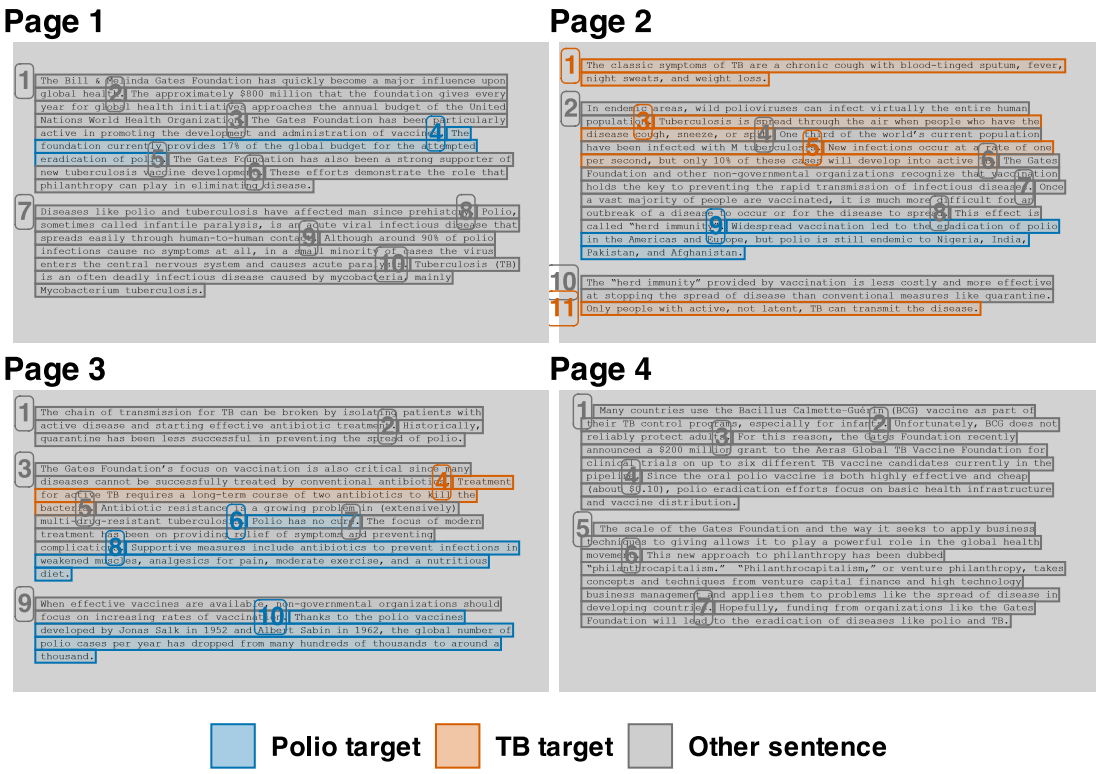
than 80 ms (4.0% of fixations) and longer than 1,000 ms (0.06%) before any further processing. Both bounds follow common practice in reading research: 80 ms is the most common lower bound, and 800, 1,000, and 1,200 ms are the most common upper bounds (Schotter & Dillon, 2025; e.g., Serrano-Carot et al., 2025). Very short fixations are unlikely to reflect processing of the text and include fixations cut short by a page change. Most fixations in reading last around 200 ms (Rayner, 1998; Schotter & Dillon, 2025); in the present data the median fixation lasted 178 ms, so a single fixation longer than 1,000 ms, 5.6 times the median, is unlikely to reflect ongoing reading and more likely reflects a lapse of attention or a brief loss of the eye-tracking signal that the fixation filter bridged.

The remaining processing (`derive_et_measures.R`) proceeded in four steps. (a) The exact layout of each page was reconstructed from the sentence texts and full-resolution screenshots of the pages (monospaced font, 79 characters per line), giving the screen position of every character of every sentence (Figure 1). (b) Each fixation was assigned to a text line after correcting for vertical drift, separately for each reader and page, with a procedure that combines two approaches used in automated drift-correction algorithms (Carr et al., 2022): assigning sequences of fixations to a line together, and modelling drift as a function of position on the page (details in Appendix D). The correction was limited to the calibration accuracy criterion of 0.5° (0.57 cm, about 0.45 of a line spacing), so it could not move a reader's gaze by a whole line. (c) Each fixation was then assigned to the sentence occupying its line and horizontal position: its character column was computed from the fixed character width, a fixation on a space took the sentence of the nearest letter, and fixations more than three characters beyond either end of a line were treated as off the text. (d) Reading on each page was taken to start at the first fixation on the page's first sentence; fixations before that (the landing) contributed to total time but not to first-pass measures. After processing, 96.8% of on-screen fixations were assigned to a sentence, and the assigned sentence agreed with Tobii Pro Lab's assignment for 89.8% of fixations. First-pass

reading was identified for 96.9% of fixated target sentences. Figure 2 shows one reader's fixations on one page before and after processing. For this reader, the correction moved fixations by a median of 0.44 line spacings and changed the sentence assigned to 43.4% of fixations; the medians over all readers and pages were 0.24 line spacings and 8.2%.

Figure 1

The Four Text Pages and the Sentence Regions Used as Areas of Interest



Note. Each sentence is one area of interest, made of one rectangle per text line it occupies. Polio target sentences are shown in blue, TB target sentences in vermillion (orange-red), and other sentences in grey; numbers give each sentence's position on its page. Page 4 contains no target sentences.

Figure 2

One Reader's Fixations on One Page Before and After Processing



Note. One participant in the TB prequestion condition, page 2, chosen to illustrate vertical drift. (A) Recorded gaze positions, coloured by the sentence assigned by Tobii Pro Lab. (B) The same fixations after drift correction (single vertical offset, assignment of chains of fixations to lines,

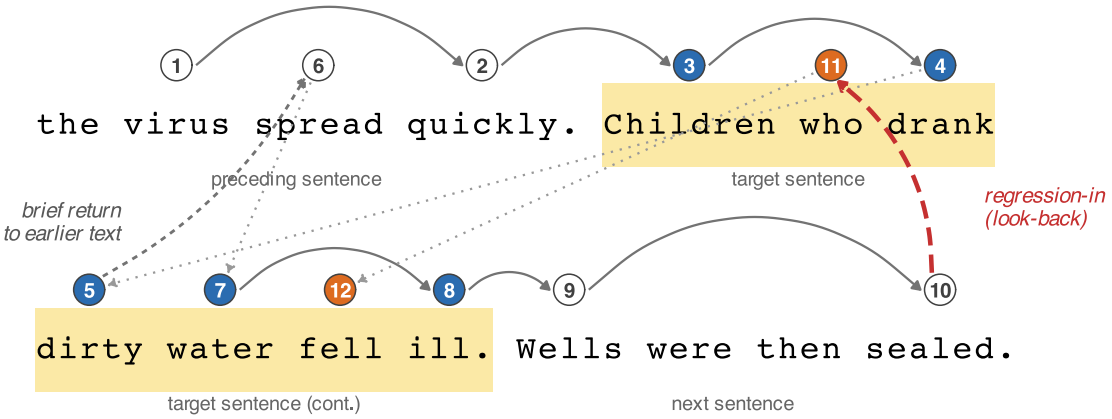
and a linear drift model, limited to 0.5°; see Appendix D), coloured by the sentence assigned in processing. Each point is one fixation (size proportional to duration); grey points were not assigned to any sentence. Lines join consecutive fixations. Open circles are fixations made before reading started on the page. Dashed rectangles outline the target sentences.

Eye-Movement Measures

All measures were computed for each participant and target sentence. Durations were divided by the number of words in the sentence (ms per word), because target sentences differed in length. Only sentences that were fixated were analysed: a target sentence that received no fixation has no reading time, so its measures were treated as missing (17 of 1380 target sentences, 1.2%). Following the eye-movement literature, we distinguish *global* measures, which reflect all processing of a sentence including later returns to it, from *local* measures, which reflect processing when the sentence is first encountered (Figure 3).

Figure 3

Eye-Movement Measures, Illustrated With a Schematic Reading Example



Measure	Type	Fixations in the example	Analysed as
Selective regression-path duration	Local	3 + 4 + 5 + 7 + 8	Ex-Gaussian (μ, τ)
Total gaze duration	Global	3 + 4 + 5 + 7 + 8 + 11 + 12	Ex-Gaussian (μ, τ)
Regression-in (look-back)	Global	1 (move from 10 to 11)	Logistic
Re-reading time	Global	11 + 12	Gamma (if re-read)
Fixation on another sentence		not counted for the target	

Note. Definitions follow Schotter and Dillon (2025). The circles are fixations, numbered in time order (arrows: saccades; dashed arrows: regressions; dotted arrows: moves to the next line); the shaded sentence is the target, and each fixation is coloured by the measure it counts towards for the target. *Selective regression-path duration* (first-pass reading; also called right-bounded reading time or selective go-past time) sums the fixations on the sentence from first entering it until the eyes first move past it to later text; fixations on earlier text during this period (here, 6) are not counted and do not end it. *Total gaze duration* sums all fixations on the sentence. *Regression-in* records whether the eyes returned to the sentence from later text (0/1). *Re-reading time* sums the fixations on the sentence after the reader had first moved on to later text (operationalizing the preregistered "re-reading durations back to the prequestioned information"). Selective regression-path duration is defined only for sentences fixated before any

later sentence (not skipped). The duration measures do not overlap: total gaze duration is the sum of selective regression-path duration, re-reading time, and any fixations made before reading started on the page. Global measures include later returns to the sentence; the local measure reflects processing when the sentence is first encountered. The preregistered first-pass duration (fixations until the sentence is first exited) would here be 3 + 4 + 5 only, cut short by the brief return at 6; it was replaced by selective regression-path duration (see Statistical Analysis). All durations were divided by the number of words in the sentence (ms per word).

Statistical Analysis

All analyses were conducted in R version 4.6.1 (R Core Team, 2026).[1] The effects of prequestions on test performance and eye movements were estimated with Bayesian mixed models in brms (Bürkner, 2017), which fits models in Stan (B. Carpenter et al., 2017), run through cmdstanr (Gabry et al., 2025) with CmdStan 2.37.0. Each target sentence, and each final-test question, was coded by its status for the reader: *prequestioned* (Polio sentences and questions for readers in the Polio condition, TB sentences and questions for readers in the TB condition), *non-prequestioned* (the other topic for those readers), or *no-prequestion* (both topics for readers in the no-prequestion condition). The main models pooled the two prequestion conditions, with Status and Topic (Polio, TB) as fixed effects. The *specific effect* of prequestions was the difference prequestioned – no-prequestion: the same sentences or questions for readers whose prequestions they answered and for readers who saw no prequestions. The *general effect* was non-prequestioned – no-prequestion. Because each topic contributes equally to the prequestioned, non-prequestioned, and no-prequestion cells, differences between the Polio and

[1] Data were processed and analysed in R version 4.6.1 (R Core Team, 2026) with the packages tidyverse 2.0.0 (Wickham et al., 2019), janitor 2.2.1 (Firke, 2024), png 0.1.9 (Urbanek, 2026), modelbased 0.16.0 (Makowski et al., 2025), marginaleffects 0.32.0 (Arel-Bundock et al., 2024), insight 1.5.2 (Lüdecke et al., 2019), brms 2.23.0 (Bürkner, 2017), cmdstanr 0.9.0 (Gabry et al., 2025), cogmod 0.3.3 (Makowski, 2026), bayestestR 0.18.1 (Makowski, Ben-Shachar, & Lüdecke, 2019), priorsense 1.4.0 (Kallioinen et al., 2023), ggdist 3.3.3 (Kay, 2024), quanteda 4.5.0 (Benoit et al., 2018), quanteda.textstats 0.97.2 (Benoit et al., 2024), udpipe 0.8.16 (Wijffels & Straka, 2026), irr 0.85 (Gamer et al., 2026), patchwork 1.3.2 (Pedersen, 2025), scales 1.4.0 (Wickham et al., 2025), knitr 1.51 (Xie, 2015), and xfun 0.60 (Xie, 2026). The manuscript was produced with Quarto 1.10.18 and the apaquarto extension.

TB materials cancel in these comparisons. Both the specific and general effects and their breakdown by topic were preregistered. To break the effects down by topic, we fitted the preregistered Group (Polio prequestions, TB prequestions, or no prequestions) × Topic (Polio, TB) models with the same families and priors. Following the preregistered hypotheses, we estimated the Polio – TB difference within each group—predicted to be positive in the Polio group, negative in the TB group, and absent in the no-prequestion group—and the Group × Topic interaction contrast, the Polio group's Polio – TB difference minus the TB group's, predicted to be positive. Because the Polio and TB sentences (and questions) are different items, each within-group Polio – TB difference combines any effect of prequestions with differences between the items, and with five items per topic, the population-level topic difference is estimated imprecisely. Each within-group difference was therefore interpreted together with the interaction contrast, in which the item differences cancel because both groups read the same items.

Reading times are strongly right-skewed, so no duration was analysed with a Gaussian model. As preregistered for heavily skewed durations, total gaze duration was analysed with an ex-Gaussian distributional model (cogmod; Makowski (2026)), which describes a duration as the sum of a Gaussian component, whose mean μ reflects typical reading time, and an exponential component, whose mean τ reflects the tail of unusually long readings. The mean of an ex-Gaussian distribution is $\mu + \tau$ and its standard deviation is $\sqrt{(\sigma^2 + \tau^2)}$, where σ is the standard deviation of the Gaussian component, so μ and τ are components of reading time rather than reading times themselves. The fixed and random effects applied to both μ and τ, and σ had by-participant random intercepts. The preregistered first-pass duration, from first entering a sentence until first leaving it, could not be described by any of the distributions we considered (ex-Gaussian, Gamma, lognormal), because it mixes two kinds of first pass. About half of first passes, in all three conditions, were cut short after a few fixations by a brief return to the preceding line, so the distribution had two modes: interrupted first passes of a few hundred

milliseconds and complete first readings of several seconds. First-pass reading was therefore measured as selective regression-path duration [also called right-bounded reading time; Schotter and Dillon (2025)]: the time spent on the sentence from first entering it until the eyes first moved past it to later text, so that brief returns to earlier text did not end it and time spent on earlier text was not counted (Figure 3). Its distribution was unimodal and right-skewed and closely described by the ex-Gaussian, so it was analysed with the same ex-Gaussian model as total gaze duration (μ and τ). Re-reading time was the time spent on a sentence after the reader had first moved on to later text, so that the two measures do not overlap (cf. rereading time in Schotter & Dillon, 2025). Whether readers returned to a sentence from later text is captured by regressions-in (0/1), which were analysed with logistic models; re-reading time was analysed for the 37.5% of sentences that were re-read, with a Gamma model with a log link, which models positive, right-skewed durations without transforming them (Lo & Andrews, 2015); the ex-Gaussian was not suitable, because re-reading times were close to exponentially distributed, so that its Gaussian component collapsed (σ near zero). The model therefore describes how long readers re-read a sentence once they had returned to it. Because the questions were worth different numbers of points (1–4), final test performance was analysed as the points earned on each question out of its maximum, with a beta-binomial model: a binomial model of the points earned with an additional parameter for overdispersion, because a participant tends to know all or none of a question's elements. Effects are reported on the scale of the expected proportion of a question's points, which puts all questions on the same 0–1 scale. Because each question has at most four points, scores can take only a few values (0, .25, .33, .50, .67, .75, and 1). In posterior predictive checks, the beta-binomial model reproduced how often each score occurred (predicted and observed shares within 1.1 percentage points), better than a binomial model without overdispersion. Prequestion performance was analysed in the same way, comparing the two prequestion conditions.

All models included the maximal random-effects structure warranted by the design (Barr et al., 2013). Topic varied within participants, so the models included by-participant random intercepts and Topic slopes (the preregistered structure). Every target sentence and every final-test question was prequestioned for one group, non-prequestioned for another, and read or answered by the no-prequestion group, so Status (and Group) varied within items, and the models also included by-sentence (eye movements) or by-question (final test) random intercepts and Status (or Group) slopes. Within a participant, Status is determined by Topic, so no by-participant Status slope was possible. In the prequestion model, condition varied between participants and between questions, so it included random intercepts for participants and questions only. Status, Condition, and Topic were coded with equal-prior contrasts [`contr.equalprior_pairs`; bayestestR, Makowski, Ben-Shachar, and Lüdecke (2019)]. With this coding, a normal(0, s) prior on the coefficients implies the same normal(0, s) prior on every pairwise difference between conditions, so that no condition serves as a reference with a different prior, and every comparison is tested against the same prior. s was set on each model's link scale: 1.5 on the log-odds scale (regressions-in, final test, and prequestions; an odds ratio of 4.5 for one *SD*), 0.5 on the log scale of the Gamma models and on the scale of τ (a factor of about 1.65 for one *SD*), and 100 ms per word for μ. Intercepts, random-effect standard deviations, and correlations had weakly informative priors (Appendix B). Each model was sampled with four chains of 10000 post-warm-up iterations (40000 posterior draws, enough for stable Bayes factors). All models converged: there were no divergent transitions, all R-hat values were below 1.001, and all bulk effective sample sizes exceeded 4900.

Means and comparisons were computed on the response scale by averaging each posterior draw's predictions over the observed sample (modelbased with the marginaleffects backend and `estimate = "average"`), so that they incorporate the variation due to the random effects. Durations are reported in ms per word, and probabilities and proportions in percent.

Effects were summarized by their posterior mean, 95% credible interval (CrI), and probability of direction (pd), the proportion of the posterior with the same sign as its median. The pd is closely related to the frequentist p value: for a one-tailed test, $p \approx 1 - \text{pd}$, and for a two-tailed test, $p \approx 2(1 - \text{pd})$, so pd > 95% corresponds to a one-tailed $p < .05$ (Makowski, Ben-Shachar, Chen, et al., 2019). The predictions for the specific effects and for prequestioned versus non-prequestioned material were directional (longer reading, more regressions, better recall), so we treated such an effect as present when pd > 95% in the predicted direction. The general effects had no directional prediction, so for them we required pd > 97.5%, corresponding to a two-tailed $p < .05$. Bayes factors were not used as a decision criterion; we report them to describe the strength of the evidence for or against each effect. Because pd only indicates the direction of an effect and cannot provide evidence that an effect is absent, we also computed Bayes factors. Each Bayes factor compares a model in which the comparison equals 0 with one in which it is free, using the Savage–Dickey density ratio (Wagenmakers et al., 2010): the posterior density of the comparison at 0 divided by its prior density at 0. The prior densities came from the same models sampled from their priors only, and both densities were computed for the comparisons on the link scale, where the equal priors apply. For τ and for the expected re-reading time, which have no single link scale, they were computed on the response scale; for the expected re-reading time, at the population level (without random effects), because averaging over random effects drawn from their priors produced prior draws too extreme for a density estimate. For the Group × Topic comparisons of look-backs and final test scores, the Bayes factors were also computed on the response scale (probabilities and proportions of points), the scale of the estimates. These comparisons involve different Polio and TB items, some of which had outcomes near 0 or 1, and for such outcomes the same comparison is much less precise in log-odds than in probability, so link-scale Bayes factors could contradict the estimates. $BF_{10} > 3$ indicates moderate, $BF_{10} > 10$ strong, $BF_{10} > 30$ very strong, and $BF_{10} > 100$ extreme evidence for a difference, and $BF_{01} = 1/BF_{10}$

the corresponding evidence for no difference; values between 1/3 and 3 are anecdotal (Lee & Wagenmakers, 2014). Because a Bayes factor depends on the prior, all comparisons between conditions received the same prior, and we report how sensitive the posteriors were to the priors [power-scaling sensitivity; Kallioinen et al. (2023)].

Deviations From the Preregistration

Deviations from the preregistration, the reasons for them, and their likely effect on the results are listed in Appendix C (Table C1).

Results

Prequestion Performance

As intended, participants knew little of the material before reading. Participants in the Polio prequestion condition earned 4.1% of the available points on the prequestions (*SD* = 5.2%) and those in the TB prequestion condition 10.5% (*SD* = 10.9%). Participants attempted 43.1% (Polio) and 39.6% (TB) of the prequestions; the rest were left blank or answered "I don't know". In a beta-binomial model of the points earned on each prequestion, with random intercepts for participants and questions, prequestion scores were similar in the two conditions (model-estimated scores: 10.6% in the TB and 8.6% in the Polio condition), TB – Polio: 2.0 percentage points, 95% CrI [–2.0, 6.0], pd = 83.0%. Because each condition answered its own five questions, this comparison cannot separate prior knowledge from the difficulty of the questions, so it is reported descriptively.

Final Test Performance

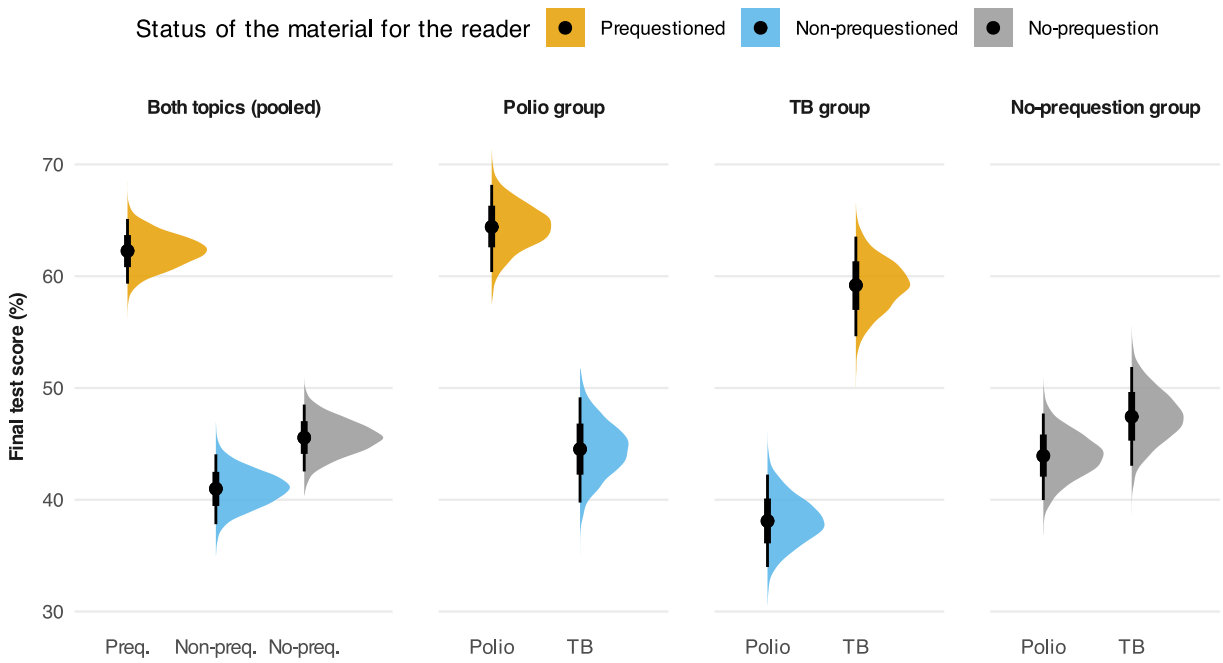
Final test performance was analysed as the proportion of each question's points earned (Figure 4). Pooled over the two prequestion conditions, participants earned 62.2% of the points on their prequestioned topic, compared with 45.6% for no-prequestion participants on the same questions, a specific benefit of 16.7 percentage points, 95% CrI [12.5, 20.7], pd > 99.9%, BF_{10} = 1.3×10^{8}: extreme evidence for a difference. On the non-prequestioned topic they earned 41.0%, numerically slightly lower than in the no-prequestion condition, –4.6 percentage points, 95% CrI

[−8.9, −0.3], pd = 98.1%, BF_{01} = 1.85. There was therefore no general benefit. Although the difference met the preregistered criterion (pd > 97.5%) in the direction of a cost, the Bayes factor was inconclusive, providing only anecdotal evidence for no difference, so we do not interpret it as a general cost.

In the Group × Topic model (Figure 4; Bayes factors on the response scale, see Statistical Analysis), the Polio group scored higher on Polio than on TB questions (64.4% vs. 44.5%), Polio − TB = 19.9 percentage points, 95% CrI [13.8, 26.0], pd > 99.9%, $BF_{10} = 8.6 \times 10^{5}$, whereas the TB group scored higher on TB than on Polio questions (59.2% vs. 38.1%), Polio − TB = −21.1 percentage points, 95% CrI [−27.1, −15.1], pd > 99.9%, $BF_{10} = 1.6 \times 10^{6}$. The no-prequestion group, which saw no prequestions, scored 43.9% on Polio and 47.5% on TB questions, Polio − TB = −3.6 percentage points, 95% CrI [−9.5, 2.3], pd = 88.3%, BF_{01} = 3.54. The Group × Topic interaction contrast—the Polio group's Polio − TB difference minus the TB group's—was 41.0 percentage points, 95% CrI [32.4, 49.4], pd > 99.9%, $BF_{10} = 1.5 \times 10^{10}$: extreme evidence that each prequestion group did relatively better on its own topic.

Figure 4

Final Test Performance: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated proportion of a question's points earned, averaged over the observed sample. Left: the status model, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion). Other panels: the Group × Topic model, for the Polio and TB questions in the Polio, TB, and no-prequestion groups; colours give the status of the questions for the readers in that group. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Eye Movements

Without prequestions, readers spent on average 249 ms per word on a target sentence in total (*SD* = 136; about 241 words per minute), 215 ms per word (*SD* = 109) of it in first-pass reading (selective regression-path duration). The ex-Gaussian estimates below divide such reading times into a typical part (μ) and a tail (τ), which sum to the mean. We first compare prequestioned and non-prequestioned sentences with the same sentences read by no-prequestion

readers (Table 1), and then test the preregistered Group × Topic crossover (Figure 5). Figure 5 shows the posterior distributions of the means for every measure, pooled and by group.

Prequestioned and Non-Prequestioned Sentences Versus the No-Prequestion Condition

Prequestions changed how readers read the sentences that answered them, compared with no-prequestion readers reading the same sentences (Table 1). During first-pass reading, measured as selective regression-path duration, typical reading time did not differ, μ: −5.0 ms per word, 95% CrI [−19.6, 9.4], pd = 75.0%, BF_{01} = 12.13, with strong evidence for no difference. The tail of unusually long first-pass readings, however, was longer for prequestioned sentences, τ: 17.1 ms per word, 95% CrI [−1.4, 36.0], pd = 96.4%, BF_{10} = 1.25, although the Bayes factor provided only anecdotal evidence for a difference. Total gaze duration showed the same pattern more clearly: prequestioned sentences had a longer tail, τ: 33.9 ms per word, 95% CrI [11.9, 56.4], pd = 99.9%, BF_{10} = 31.07, very strong evidence for a difference, whereas typical reading time did not differ, μ: 2.8 ms per word, 95% CrI [−11.4, 17.0], pd = 64.7%, BF_{01} = 14.26, with strong evidence for no difference. After moving on, readers were more likely to return to prequestioned sentences from later text (37.6% vs. 30.2%), 7.3 percentage points, 95% CrI [2.4, 12.2], pd = 99.8%, BF_{10} = 1.67, anecdotal evidence, and when they returned, they re-read prequestioned sentences for longer (129.2 ms per word vs. 100.0 ms per word), 29.2 ms per word, 95% CrI [7.8, 51.2], pd = 99.6%, BF_{10} = 9.49, moderate evidence.

Non-prequestioned sentences, by contrast, were read like the same sentences in the no-prequestion condition. No measure showed a general effect (non-prequestioned − no-prequestion, Table 1; all pd < 97.5%), and on every measure the Bayes factor favoured its absence (BF_{01} from 3.7 to 15.4), with moderate or stronger evidence on every measure. The effects of prequestions on reading were therefore specific to the prequestioned sentences.

Table 1

Prequestioned and Non-Prequestioned Sentences Versus the Same Sentences Read Without Prequestions

Measure	Prequestioned – No-prequestion	pd	BF_{10}	Non-prequestioned – No-prequestion	pd	BF_{01}
Selective regression path: μ	–5.0 [–19.6, 9.4]	75.0%	0.08	–4.9 [–19.0, 9.2]	75.5%	12.27
Selective regression path: τ	17.1 [–1.4, 36.0]	96.4%	1.25	–0.1 [–17.1, 17.1]	50.3%	4.62
Total gaze: μ	2.8 [–11.4, 17.0]	64.7%	0.07	2.3 [–11.2, 15.9]	63.1%	15.41
Total gaze: τ	33.9 [11.9, 56.4]	99.9%	31.07	–4.7 [–24.0, 14.7]	68.5%	3.72
Regression-in	7.3 [2.4, 12.2]	99.8%	1.67	–1.3 [–6.1, 3.5]	70.6%	6.12
Re-reading time	29.2 [7.8, 51.2]	99.6%	9.49	0.3 [–19.8, 20.6]	51.2%	5.56

Note. Posterior means and 95% credible intervals of the differences, pooled over the Polio and TB prequestion conditions and averaged over the two topics, in ms per word (durations) or percentage points (regressions-in). Prequestioned – no-prequestion is the specific effect and non-prequestioned – no-prequestion the general effect of prequestions. Selective regression-path duration (first-pass reading) and total gaze duration were analysed with ex-Gaussian models (μ = typical reading time; τ = tail of unusually long readings); regressions-in with a logistic model; re-reading time, for sentences that were re-read, with a Gamma model. pd = probability of direction. BF_{10} = Bayes factor for a difference (Savage–Dickey); $BF_{01} = 1/BF_{10}$, evidence for no difference. Selective regression-path duration is a local measure; all others are global.

Group × Topic

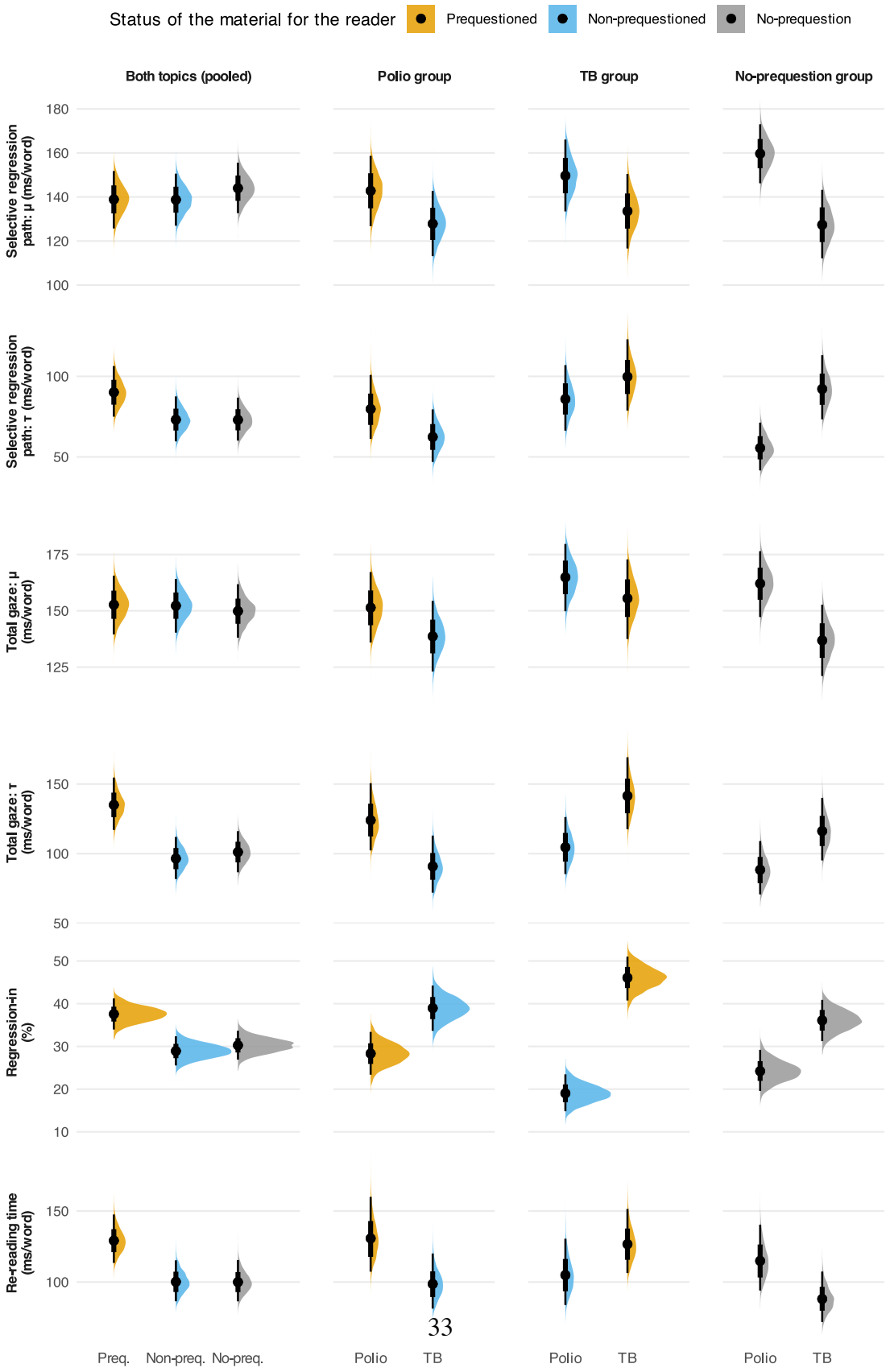
The preregistered prediction was a crossover: longer reading of Polio than of TB sentences in the Polio group, the reverse in the TB group, and no difference in the no-prequestion group, so that the Group × Topic interaction contrast would be positive. The interaction contrast

was positive (pd > 95%) for the tail of first-pass reading (τ): 31.1 ms per word, 95% CrI [–3.4, 66.4], pd = 96.1%, BF_{10} = 1.68; the tail of total reading time (τ): 70.1 ms per word, 95% CrI [29.2, 112.0], pd = 99.9%, BF_{10} = 109; look-backs from later text: 16.5 percentage points, 95% CrI [6.9, 26.0], pd > 99.9%, BF_{10} = 63.06; and re-reading time: 54.0 ms per word, 95% CrI [11.0, 97.4], pd = 99.3%, BF_{10} = 4.79. It was not reliably positive for typical first-pass reading time (μ): –1.1 ms per word, 95% CrI [–27.6, 25.6], pd = 53.5%, BF_{01} = 9.40 and typical total reading time (μ): 3.5 ms per word, 95% CrI [–23.4, 30.5], pd = 60.1%, BF_{01} = 9.06. For the tail of total gaze duration (τ), the Polio – TB difference was 33.2 ms per word, 95% CrI [4.6, 62.7], pd = 98.9%, BF_{10} = 4.24 in the Polio group, –36.9 ms per word, 95% CrI [–68.6, –6.1], pd = 99.0%, BF_{10} = 5.07 in the TB group, and –27.9 ms per word, 95% CrI [–55.9, –1.0], pd = 97.9%, BF_{10} = 2.27 in the no-prequestion group. For look-backs from later text (Bayes factors on the response scale), it was –10.6 percentage points, 95% CrI [–17.7, –3.4], pd = 99.8%, BF_{10} = 10.73 in the Polio group, –27.2 percentage points, 95% CrI [–33.7, –20.5], pd > 99.9%, $BF_{10} = 2.2 \times 10^{7}$ in the TB group, and –11.9 percentage points, 95% CrI [–18.6, –5.1], pd > 99.9%, BF_{10} = 58.69 in the no-prequestion group. For re-reading time, it was 32.2 ms per word, 95% CrI [1.5, 64.5], pd = 98.0%, BF_{10} = 2.99 in the Polio group, –21.8 ms per word, 95% CrI [–54.1, 11.3], pd = 90.7%, BF_{01} = 3.64 in the TB group, and 26.6 ms per word, 95% CrI [–1.5, 56.6], pd = 96.8%, BF_{10} = 2.51 in the no-prequestion group. Because the Polio and TB sentences differ, the no-prequestion group shows how the topics are read without prequestions (Figure 5). Compared with that group, the crossover in look-backs came mainly from the TB group, who looked back to TB sentences more often (46.1% vs. 36.1%) and to Polio sentences less often (19.0% vs. 24.2%) than no-prequestion readers, whereas the Polio group's look-backs differed little from those of no-prequestion readers on either topic (Polio: 28.4% vs. 24.2%; TB: 39.0% vs. 36.1%). Re-reading time showed the same asymmetry: the TB group re-read TB sentences for longer than no-prequestion readers (126.9 ms per word vs. 88.1 ms per word), whereas the Polio group's re-

reading of Polio sentences was closer to that of no-prequestion readers (130.8 ms per word vs. 114.7 ms per word). These comparisons across groups are descriptive.

Figure 5

Eye-Movement Measures: Posterior Distributions of the Model-Estimated Means



Note. Posterior distributions of the model-estimated means, averaged over the observed sample. For the ex-Gaussian measures, μ and τ are shown separately; a mean reading time is μ + τ. Left: the status models, pooled over the two prequestion groups and both topics (prequestioned, non-prequestioned, and no-prequestion sentences). Other panels: the Group × Topic models, for the Polio and TB sentences in the Polio, TB, and no-prequestion groups; colours give the status of the sentences for the readers in that group. μ and τ are the Gaussian mean (typical reading time) and the tail mean (unusually long readings) of the ex-Gaussian models of selective regression-path duration (first-pass reading) and total gaze duration. Durations are in ms per word; regressions-in are the percentage of sentences returned to from later text; re-reading time is for sentences that were re-read. Preq. = prequestioned; non-preq. = non-prequestioned; no-preq. = no-prequestion. Points are posterior means; thick and thin bars are 66% and 95% credible intervals.

Discussion

This study used eye tracking to test whether prequestions improve memory by directing attention toward the information they address. On the final test, prequestions produced the typical specific benefit and no general benefit, and each prequestion group did relatively better on its own topic (interaction contrast, extreme evidence). Eye movements showed the same specificity. Compared with the same sentences read without prequestions, prequestioned sentences received more unusually long readings, already during first-pass reading, were looked back to more often from later text, and were re-read for longer when readers returned to them; non-prequestioned sentences were read as they were without prequestions on every measure. In the Group × Topic comparisons, the interaction contrast was positive (pd > 95%) for the tail of first-pass reading (τ), the tail of total reading time (τ), look-backs from later text, and re-reading time; and not reliably positive for typical first-pass reading time (μ) and typical total reading time (μ) (Figure 5).

These effects show what kind of attention prequestions recruit. Prequestions did not slow reading of the relevant material overall: the Bayes factors favoured no difference in typical reading time (μ), both in first-pass reading and in total gaze duration. Instead, the additional processing appeared on a subset of encounters, as occasional unusually long readings (τ), and in returns to the answers. It was also confined to the prequestioned sentences, so prequestions changed how readers treated their answers, not how they read the passage. This fits the absence of both a general benefit and a general cost in the meta-analysis of King-Shepard et al. (2025), and it argues against accounts on which prequestions raise curiosity or monitoring for the topic as a whole.

We suggest that this additional processing is triggered when a reader recognizes that a sentence answers one of the prequestions. Lewis and Mensink (2012) proposed a similar mechanism: a sentence that shares words and meaning with a prequestion "resonates" with the reader's memory of the question and prompts the reader to re-inspect it. Recognition can occur only once enough of a sentence has been read, and it can fail if the reader misses the connection or has forgotten the question, so it should lengthen some readings rather than shift all of them—an effect on τ but not μ. Unlike text degradation, which slows all fixations (White & Staub, 2012), prequestions lengthened only a subset of readings, as familiarity does in scene viewing (Guy et al., 2020). The longer tail in first-pass reading suggests that recognition often occurs as an answer is first read; the look-backs and longer re-reading suggest that readers then return to the answer, perhaps to compare it with the question or their earlier guess, or to rehearse it. Recognition may also be what makes prequestions effective: in video lectures, prequestions enhanced learning only when learners remembered them (St. Hilaire & Carpenter, 2020). The attentional-window account (Sana & Carpenter, 2023) explains the long readings—attention is heightened until an answer is found—but not the look-backs, because once an answer has been found attention should return to its usual level.

The results also add to evidence that attention is not the whole story. If prequestions simply heightened attention to the content they address, that content should have been read more slowly throughout, shifting μ. Instead, processing increased only on some encounters with an answer and in later returns to it, consistent with findings that prequestions can improve learning without increasing study time (Motz et al., 2026). This sequence—recognition first, attention after—is what the mental model hypothesis predicts (Motz et al., 2026), although our non-prequestioned sentences, about the other disease, cannot show whether activated knowledge also helps related information on the prequestioned topic. The pauses and returns also resemble metacognitive monitoring, in which learners look for information that fills the gaps the prequestions revealed (King-Shepard et al., 2025); recognizing an answer is itself a metacognitive judgment. Eye movements alone cannot separate these accounts, but combining them with direct measures of metacognition, such as judgments of learning (King-Shepard et al., 2025), could. Nor can these data show that the additional processing produces the memory benefit: reading was observed rather than manipulated, and prequestions may also strengthen memory in ways that leave little trace in reading time. Experiments that control whether readers can return to the answers could test this.

These findings agree with the two previous eye-tracking studies in showing that prequestions shift attention toward the information they address. Because our TB group read the same text as Experiment 2 of Lewis and Mensink (2012) after prequestions on the same topic, comparing it with the no-prequestion group on the TB material is close to a replication of that experiment (descriptive comparisons from the Group × Topic models). As in Lewis and Mensink (2012), TB prequestions improved memory for the TB information (11.8 percentage points, 95% CrI [5.3, 17.9]) and increased looking back to the TB sentences from later text (10.0 percentage points, 95% CrI [3.0, 16.9]) and the time spent re-reading them (38.6 ms per word, 95% CrI [12.2, 67.1]). Unlike Lewis and Mensink (2012), who found more re-inspection of the targeted

sentences during first-pass reading, we found no reliable difference in first-pass reading of the TB sentences (μ: 6.2 ms per word, 95% CrI [−13.9, 26.9]; τ: 7.6 ms per word, 95% CrI [−19.6, 35.3]). Pooled over both prequestion groups, however, selective regression-path duration showed a longer tail but no shift in typical reading time, although with weak evidence. Separating μ from τ was important here: an analysis of means, as in previous studies, would dilute effects confined to a subset of readings and could not distinguish them from slower reading overall. The studies also differ in design and measures: the present study compared every target sentence with and without prequestions in a larger sample, measured re-reading as time on the sentence itself after the reader had moved on, and measured first-pass reading as selective regression-path duration (Schotter & Dillon, 2025), because about half of first passes were cut short by a brief return to the preceding line. Bostan and Ozcelik (2026) likewise found more fixations on prequestioned sentences and more transitions between them, paralleling our look-backs, but unlike most prequestion research they also found a general benefit. Their integrative prequestions required combining information across the text and may have directed attention to non-prequestioned information as well (Pan & Carpenter, 2023), whereas our prequestions about single facts did neither. With 12 participants per group, however, their estimates are imprecise.

Several limitations should be noted. Each topic was represented by only five target sentences and five test questions, and the Polio and TB sentences differed in length, content, reading level, and position on the page, so the within-group Polio − TB differences combine effects of prequestions with differences between the materials. We therefore based our conclusions on the pooled comparisons, which compare the same sentences and questions with and without prequestions, and treated the Group × Topic contrasts as converging evidence. The topics also differed in how strongly prequestions affected them: the increase in look-backs and re-reading came mainly from the TB group, whose target sentences were already looked back to more often without prequestions. Future studies should control target sentences more closely—

matching them across topics in length, reading level, position on the page, and wording overlap with the questions (Lewis & Mensink, 2012), or varying which sentences serve as targets across versions of the text—and use more of them. The comparisons with the no-prequestion condition are between participants and so less precise than within-reader comparisons. The Bayes factors depend on the priors: they show that the data are better described without a general effect than with one of the size the priors allow, not that a small general effect is impossible. Finally, eye movements were recorded without head stabilization, and vertical drift was corrected only within the calibration accuracy criterion.

In sum, prequestions guide where readers look, and only toward the information they target: rather than reading the relevant material more slowly, readers pause on some of the answers and return to them later, while reading the rest of the text as they would without prequestions. Prequestions therefore do not simply heighten attention to the material they address. Such recognition-triggered processing is a plausible route by which prequestions strengthen memory for their answers, but whether it accounts for the memory benefit remains to be tested.

References

Arel-Bundock, V., Greifer, N., & Heiss, A. (2024). How to interpret statistical models using marginaleffects for R and Python. *Journal of Statistical Software*, *111*(9), 1–32. https://doi.org/10.18637/jss.v111.i09

Ashby, J., Rayner, K., & Clifton, C. (2005). Eye movements of highly skilled and average readers: Differential effects of frequency and predictability. *The Quarterly Journal of Experimental Psychology Section A*, *58*(6), 1065–1086. https://doi.org/10.1080/02724980443000476

Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry: The power of response time distributional analyses. *Current Directions in Psychological Science*, *20*(3), 160–166. https://doi.org/10.1177/0963721411408885

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Benoit, K., Watanabe, K., Wang, H., Lua, J. W., & Kuha, J. (2024). *Quanteda.textstats: Textual statistics for the quantitative analysis of textual data*. https://quanteda.io

Benoit, K., Watanabe, K., Wang, H., Nulty, P., Obeng, A., Müller, S., & Matsuo, A. (2018). Quanteda: An r package for the quantitative analysis of textual data. *Journal of Open Source Software*, *3*(30), 774. https://doi.org/10.21105/joss.00774

Berlyne, D. E. (1954). A theory of human curiosity. *British Journal of Psychology*, *45*(3), 180–191. https://doi.org/10.1111/j.2044-8295.1954.tb01243.x

Berlyne, D. E. (1962). Uncertainty and epistemic curiosity. *British Journal of Psychology*, *53*(1), 27–34. https://doi.org/10.1111/j.2044-8295.1962.tb00811.x

Bjork, R. A. (1994). Memory and metamemory considerations in the training of human beings. In J. Metcalfe & A. P. Shimamura (Eds.), *Metacognition: Knowing about knowing* (pp. 185–205). MIT Press.

Bjork, R. A., Dunlosky, J., & Kornell, N. (2013). Self-regulated learning: Beliefs, techniques, and illusions. *Annual Review of Psychology*, *64*, 417–444. https://doi.org/10.1146/annurev-psych-113011-143823

Bostan, E., & Ozcelik, E. (2026). The effect of integrative prequestions on learning from text: An eye-tracking study. *The Journal of Experimental Education*, *94*(1), 33–45. https://doi.org/10.1080/00220973.2024.2355980

Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Carpenter, S. K. (2012). Testing enhances the transfer of learning. *Current Directions in Psychological Science*, *21*(5), 279–283. https://doi.org/10.1177/0963721412452728

Carpenter, S. K., King-Shepard, Q., & Nokes-Malach, T. J. (2023). The prequestion effect: Why it is useful to ask students questions before they learn. In C. E. Overson, C. M. Hakala, L. L. Kordonowy, & V. A. Benassi (Eds.), *In their own words: What scholars want you to know about why and how to apply the science of learning in your academic setting* (pp. 74–82). Society for the Teaching of Psychology.

Carpenter, S. K., Rahman, S., & Perkins, K. (2018). The effects of prequestions on classroom learning. *Journal of Experimental Psychology: Applied*, *24*(1), 34–42. https://doi.org/10.1037/xap0000145

Carpenter, S. K., & Toftness, A. R. (2017). The effect of prequestions on learning from video presentations. *Journal of Applied Research in Memory and Cognition*, *6*(1), 104–109. https://doi.org/10.1016/j.jarmac.2016.07.014

Carr, J. W., Pescuma, V. N., Furlan, M., Ktori, M., & Crepaldi, D. (2022). Algorithms for the automated correction of vertical drift in eye-tracking data. *Behavior Research Methods*, *54*(1), 287–310. https://doi.org/10.3758/s13428-021-01554-0

Daneman, M., & Carpenter, P. A. (1980). Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal Behavior*, *19*(4), 450–466. https://doi.org/10.1016/S0022-5371(80)90312-6

Deng, R., & Gao, Y. (2023). A review of eye tracking research on video-based learning. *Education and Information Technologies*, *28*(6), 7671–7702. https://doi.org/10.1007/s10639-022-11486-7

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, *14*(1), 4–58. https://doi.org/10.1177/1529100612453266

Firke, S. (2024). *Janitor: Simple tools for examining and cleaning dirty data*. https://github.com/sfirke/janitor

Gabry, J., Češnovar, R., Johnson, A., & Bronder, S. (2025). *Cmdstanr: R interface to CmdStan*. https://mc-stan.org/cmdstanr/

Gamer, M., Lemon, J., Fellows, I., & Singh, P. (2026). *Irr: Various coefficients of interrater reliability and agreement*. https://www.r-project.org

Guy, N., Lancry-Dayan, O. C., & Pertzov, Y. (2020). Not all fixations are created equal: The benefits of using ex-Gaussian modeling of fixation durations. *Journal of Vision*, *20*(10), 9. https://doi.org/10.1167/jov.20.10.9

Hartwig, M. K., & Dunlosky, J. (2012). Study strategies of college students: Are self-testing and scheduling related to achievement? *Psychonomic Bulletin & Review*, *19*(1), 126–134. https://doi.org/10.3758/s13423-011-0181-y

Hyönä, J., Radach, R., & Deubel, H. (Eds.). (2003). *The mind's eye: Cognitive and applied aspects of eye movement research*. Elsevier.

Kallioinen, N., Paananen, T., Bürkner, P.-C., & Vehtari, A. (2023). Detecting and diagnosing prior and likelihood sensitivity with power-scaling. *Statistics and Computing*, *34*. https://doi.org/10.1007/s11222-023-10366-5

Kay, M. (2024). ggdist: Visualizations of distributions and uncertainty in the grammar of graphics. *IEEE Transactions on Visualization and Computer Graphics*, *30*(1), 414–424. https://doi.org/10.1109/TVCG.2023.3327195

Kincaid, J. P., Fishburne, R. P., Rogers, R. L., & Chissom, B. S. (1975). *Derivation of new readability formulas (automated readability index, fog count and Flesch reading ease formula) for Navy enlisted personnel* (Nos. Research Branch Report 8-75). Naval Technical Training Command.

King-Shepard, Q. W., Walker, J., Nokes-Malach, T. J., Carpenter, S. K., & Fraundorf, S. H. (2025). The effect of prequestions on learning: A multilevel meta-analysis. *Educational Psychology Review*, *37*, 115. https://doi.org/10.1007/s10648-025-10075-7

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, *15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Kornell, N., Hays, M. J., & Bjork, R. A. (2009). Unsuccessful retrieval attempts enhance subsequent learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *35*(4), 989–998. https://doi.org/10.1037/a0015729

Kornell, N., & Vaughn, K. E. (2016). How retrieval attempts affect learning: A review and synthesis. In B. H. Ross (Ed.), *Psychology of learning and motivation* (Vol. 65, pp. 183–215). Academic Press. https://doi.org/10.1016/bs.plm.2016.03.003

Kumle, L., Võ, M. L.-H., & Draschkow, D. (2021). Estimating power in (generalized) linear mixed models: An open introduction and tutorial in R. *Behavior Research Methods*, *53*(6), 2528–2543. https://doi.org/10.3758/s13428-021-01546-0

Lee, M. D., & Wagenmakers, E.-J. (2014). *Bayesian cognitive modeling: A practical course*. Cambridge University Press. https://doi.org/10.1017/CBO9781139087759

Lewis, M. R., & Mensink, M. C. (2012). Prereading questions and online text comprehension. *Discourse Processes*, *49*(5), 367–390. https://doi.org/10.1080/0163853X.2012.662801

Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, *9*(2), 159–191.

Lo, S., & Andrews, S. (2015). To transform or not to transform: Using generalized linear mixed models to analyse reaction time data. *Frontiers in Psychology*, *6*, 1171. https://doi.org/10.3389/fpsyg.2015.01171

Lüdecke, D., Waggoner, P., & Makowski, D. (2019). insight: A unified interface to access information from model objects in R. *Journal of Open Source Software*, *4*(38), 1412. https://doi.org/10.21105/joss.01412

Madsen, J., Júlio, S. U., Gucik, P. J., Steinberg, R., & Parra, L. C. (2021). Synchronized eye movements predict test scores in online video education. *Proceedings of the National Academy of Sciences*, *118*(5), e2016980118. https://doi.org/10.1073/pnas.2016980118

Makowski, D. (2026). *Cogmod: Cognitive models for subjective scales and decision making tasks*. https://github.com/DominiqueMakowski/cogmod

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the Bayesian framework. *Frontiers in Psychology*, *10*, 2767. https://doi.org/10.3389/fpsyg.2019.02767

Makowski, D., Ben-Shachar, M. S., & Lüdecke, D. (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the bayesian framework. *Journal of Open Source Software*, *4*(40), 1541. https://doi.org/10.21105/joss.01541

Makowski, D., Ben-Shachar, M. S., Wiernik, B. M., Patil, I., Thériault, R., & Lüdecke, D. (2025). modelbased: An R package to make the most out of your statistical models through marginal means, marginal effects, and model predictions. *Journal of Open Source Software*, *10*(109), 7969. https://doi.org/10.21105/joss.07969

Masked Citation. (n.d.). *Masked Title*.

Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-Gaussian and shifted Wald parameters: A diffusion model analysis. *Psychonomic Bulletin & Review*, *16*(5), 798–817. https://doi.org/10.3758/PBR.16.5.798

McNamara, D. S., & Magliano, J. (2009). Toward a comprehensive model of comprehension. In B. H. Ross (Ed.), *The psychology of learning and motivation* (Vol. 51, pp. 297–384). Elsevier. https://doi.org/10.1016/S0079-7421(09)51009-2

Motz, B. A., Chinni, A., de Leeuw, J. R., Jankowski, H. E., Aggarwal, A., Amato, M., Berlin, K., Britten, K., Brown, A., Cerchiaro, M. L., Evans, N. J., Findley, A., Gorman, R., Gregg, K. D., Hansen, K., Hollender, H., Hullinger, R. A., Larkin, P., Lion, M., … Fyfe, E. R. (2026). ManyClasses 2: The effects of prequestions on media interactions and learning. *Journal of Educational Psychology*, *118*(2), 153–175. https://doi.org/10.1037/edu0000978

Olsen, A. (2012). *The Tobii I-VT fixation filter: Algorithm description*. Tobii Technology.

Pan, S. C., & Carpenter, S. K. (2023). Prequestioning and pretesting effects: A review of empirical research, theoretical perspectives, and implications for educational practice. *Educational Psychology Review*, *35*, 97. https://doi.org/10.1007/s10648-023-09814-5

Pedersen, T. L. (2025). *Patchwork: The composer of plots*. https://patchwork.data-imaginist.com

Perfetti, C. (2007). Reading ability: Lexical quality to comprehension. *Scientific Studies of Reading*, *11*(4), 357–383. https://doi.org/10.1080/10888430701530730

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://doi.org/10.32614/R.manuals

Rayner, K. (1998). Eye movements in reading and information processing: 20 years of research. *Psychological Bulletin*, *124*(3), 372–422. https://doi.org/10.1037/0033-2909.124.3.372

Rayner, K. (2009). Eye movements and attention in reading, scene perception, and visual search. *Quarterly Journal of Experimental Psychology*, *62*(8), 1457–1506. https://doi.org/10.1080/17470210902816461

Richland, L. E., Kornell, N., & Kao, L. S. (2009). The pretesting effect: Do unsuccessful retrieval attempts enhance learning? *Journal of Experimental Psychology: Applied*, *15*(3), 243–257. https://doi.org/10.1037/a0016496

Rivers, M. L., Berdelis, A. J., & Pan, S. C. (2026). Does covert prequestioning enhance learning? Implications for the roles of attention and retrieval in the prequestioning effect. *Applied Cognitive Psychology*. https://doi.org/10.1002/acp.70165

Roediger, H. L., & Karpicke, J. D. (2006). The power of testing memory: Basic research and implications for educational practice. *Perspectives on Psychological Science*, *1*(3), 181–210. https://doi.org/10.1111/j.1745-6916.2006.00012.x

Rothkopf, E. Z., & Billington, M. J. (1979). Goal-guided learning from text: Inferring a descriptive processing model from inspection times and eye movements. *Journal of Educational Psychology*, *71*(3), 310–327. https://doi.org/10.1037/0022-0663.71.3.310

Rowland, C. A. (2014). The effect of testing versus restudy on retention: A meta-analytic review of the testing effect. *Psychological Bulletin*, *140*(6), 1432–1463. https://doi.org/10.1037/a0037559

Salvucci, D. D., & Goldberg, J. H. (2000). Identifying fixations and saccades in eye-tracking protocols. *Proceedings of the 2000 Symposium on Eye Tracking Research & Applications*, 71–78. https://doi.org/10.1145/355017.355028

Sana, F., & Carpenter, S. K. (2023). Broader benefits of the pretesting effect: Placement matters. *Psychonomic Bulletin & Review*. https://doi.org/10.3758/s13423-023-02274-6

Sana, F., Forrin, N. D., Sharma, M., Dubljevic, T., Ho, P., Jalil, E., & Kim, J. A. (2020). Optimizing the efficacy of learning objectives through pretests. *CBE—Life Sciences Education*, *19*(3), ar43. https://doi.org/10.1187/cbe.19-11-0257

Schotter, E. R., & Dillon, B. (2025). A beginner's guide to eye tracking for psycholinguistic studies of reading. *Behavior Research Methods*, *57*(2), 68. https://doi.org/10.3758/s13428-024-02572-4

Serrano-Carot, M., Angele, B., Xu, H., & Vasilev, M. R. (2025). *Webcams can be used to study eye movements during reading*. PsyArXiv preprint; PsyArXiv. https://doi.org/10.31234/osf.io/bzt2h_v1

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, *86*(2), 420–428. https://doi.org/10.1037/0033-2909.86.2.420

Silveira, N., Dozat, T., Marneffe, M.-C. de, Bowman, S., Connor, M., Bauer, J., & Manning, C. (2014). A gold standard dependency corpus for English. *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, 2897–2904.

Soderstrom, N. C., & Bjork, E. L. (2023). Pretesting enhances learning in the classroom. *Educational Psychology Review*, *35*(3), 88. https://doi.org/10.1007/s10648-023-09805-6

St. Hilaire, K. J., & Carpenter, S. K. (2020). Prequestions enhance learning, but only when they are remembered. *Journal of Experimental Psychology: Applied*, *26*(4), 705–716. https://doi.org/10.1037/xap0000296

St. Hilaire, K. J., Carpenter, S. K., & Jennings, J. M. (2019). Using prequestions to enhance learning from reading passages: The roles of question type and structure building ability. *Memory*, *27*(9), 1204–1213. https://doi.org/10.1080/09658211.2019.1641209

St. Hilaire, K. J., Chan, J. C. K., & Ahn, D. (2024). Guessing as a learning intervention: A meta-analytic review of the prequestion effect. *Psychonomic Bulletin & Review*, *31*(2), 411–441. https://doi.org/10.3758/s13423-023-02353-8

Staub, A., White, S. J., Drieghe, D., Hollway, E. C., & Rayner, K. (2010). Distributional effects of word frequency on eye fixation durations. *Journal of Experimental Psychology: Human Perception and Performance*, *36*(5), 1280–1293. https://doi.org/10.1037/a0016896

Straka, M., & Straková, J. (2017). Tokenizing, POS tagging, lemmatizing and parsing UD 2.0 with UDPipe. *Proceedings of the CoNLL 2017 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, 88–99. https://doi.org/10.18653/v1/K17-3009

Urbanek, S. (2026). *Png: Read and write PNG images*. https://www.rforge.net/png/

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage–Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

White, S. J., & Staub, A. (2012). The distribution of fixation durations during reading: Effects of stimulus quality. *Journal of Experimental Psychology: Human Perception and Performance*, *38*(3), 603–617. https://doi.org/10.1037/a0025338

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M.,

Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., Pedersen, T. L., & Seidel, D. (2025). *Scales: Scale functions for visualization*. https://scales.r-lib.org

Wijffels, J., & Straka, M. (2026). *Udpipe: Tokenization, parts of speech tagging, lemmatization and dependency parsing with the UDPipe 'NLP' toolkit*. https://bnosac.github.io/udpipe/en/index.html

Xie, Y. (2015). *Dynamic documents with R and knitr* (2nd ed.). Chapman; Hall/CRC. https://yihui.org/knitr/

Xie, Y. (2026). *Xfun: Supporting functions for packages maintained by yihui xie*. https://github.com/yihui/xfun

Yang, J., Zhang, Y., Pi, Z., & Xie, Y. (2021). Students' achievement motivation moderates the effects of interpolated pre-questions on attention and learning from video lectures. *Learning and Individual Differences*, *91*, 102055. https://doi.org/10.1016/j.lindif.2021.102055

Appendix A

Questions and Scoring Rules

Prequestion and final test responses were scored with the same rule-based script (`scoring_rubric.R`), written by Claude Opus 5.5 (Anthropic; model ID `claude-opus-5-5`) and applied identically to every response. Each question awarded one point per correct element present in the response (Table A1, Table A2). Blank responses, "I don't know" and similar responses, and random keystrokes scored 0. Minor misspellings were accepted when the intended word was unambiguous (e.g., "afganistan"). The same rules were used for the prequestions and the final test.

Table A1

Final Test Questions and Answers

Q	Topic	Question	Answer	Points
1	Polio	What percent of the global budget does the Gates Foundation devote toward the eradication of polio	17%	1
2	Polio	Is there a cure for polio?	no	1
3	Polio	In which countries is polio still a major concern?	Nigeria, India, Pakistan, and Afghanistan	4
4	Polio	What measures are included in the modern treatment of polio?	antibiotics to prevent infections in weakened muscles, analgesics for pain, moderate exercise, and a nutritious diet	4
5	TB	What are the symptoms of tuberculosis?	a chronic cough with blood-tinged sputum, fever, night sweats, and weight loss	4
6	TB	What percentage of new infections will turn into active cases of tuberculosis?	10%	1
7	TB	What is the treatment for tuberculosis?	a long-term course of two antibiotics to kill the bacteria.	3
8	TB	Can people with latent tuberculosis spread the disease?	no	1
9	TB	How is tuberculosis spread through the air?	cough, sneeze, or spit	3
10	Polio	Since the development of the polio vaccine, the global number of polio cases per year has dropped from _____ to _____.	many hundreds of thousands to around a thousand	2

Note. Points = maximum points for the question. The Polio prequestion condition answered questions 1–4 and 10 before reading; the TB prequestion condition answered questions 5–9.

Table A2

Scoring Rules and Agreement With the Second Rater for Each Question

Q	Credited elements	*n*	Exact	ICC
1	The number 17 (as a whole number, e.g., not 170).	158	100.0%	1.00
2	A negative answer: “no”, “no cure”, or “not a cure”.	198	100.0%	1.00
3	One point each for Nigeria, India, Pakistan, and Afghanistan.	164	97.0%	.99
4	One point each for antibiotics; analgesics or pain relief; exercise (including physical activity or therapy); and diet or nutrition. “Antibiotic resistance” alone was not credited.	162	98.8%	.99
5	One point each for cough (or blood-tinged sputum/ phlegm); fever (or temperature); sweats (including night sweats); and weight loss.	180	91.7%	.96
6	10% (or “ten percent”, 1/10); not credited if stated as “less than” or “under” 10%.	155	99.4%	.99
7	One point each for antibiotics (or antibacterials); two (two drugs, courses, rounds, or types); and long-term (long, extended, prolonged, or months).	168	97.6%	.98
8	A negative answer (“no”).	185	100.0%	1.00
9	One point each for cough, sneeze, and spit (or saliva).	174	99.4%	1.00
10	One point each for a starting value of hundreds of thousands (100,000–999,999) and an end value of about a thousand (500–9,999, or “a thousand”).	134	91.0%	.94

Note. One point was awarded for each credited element present in a response, up to the question’s maximum (see Table A1). *n* = number of non-blank responses (prequestions and final

test) scored by both the automated rubric and the second rater; Exact = percentage of responses given the same score; ICC = ICC(2,1), absolute agreement.

Appendix B

Model Specifications

Table B1

Bayesian Models: Families, Effects, and Priors

Measure	Family	Random effects	Prior on comparisons
Total gaze duration	Ex-Gaussian (μ identity; σ, τ softplus)	μ and τ: participants (intercept, Topic slope), sentences (intercept, Status or Group slope); σ: participants (intercept)	μ: normal(0, 0.1) s/word; τ: normal(0, 0.5)
Selective regression-path duration	Ex-Gaussian (μ identity; σ, τ softplus)	μ and τ: participants (intercept, Topic slope), sentences (intercept, Status or Group slope); σ: participants (intercept)	μ: normal(0, 0.1) s/word; τ: normal(0, 0.5)
Re-reading time	Gamma (log)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 0.5)
Regression-in (look-back)	Bernoulli (logit)	Participants (intercept, Topic slope), sentences (intercept, Status or Group slope)	normal(0, 1.5)
Final test	Beta-binomial (logit)	Participants (intercept, Topic slope), questions (intercept, Status or Group slope)	normal(0, 1.5)
Prequestions	Beta-binomial (logit)	Participants (intercept), questions (intercept)	normal(0, 1.5)

Note. Pooled models: Status (prequestioned, non-prequestioned, no-prequestion) + Topic; Group × Topic models: Group (Polio, TB, no-prequestion) × Topic. Status, Condition, and Topic were coded with equal-prior contrasts, so the prior on the coefficients is also the prior on every pairwise difference between conditions (on the link scale). Other priors: intercepts normal(−1.5, 1) on the log scale (Gamma models; about 0.2 s per word), normal(0, 1.5) on the log-odds scale, and normal(0.2, 0.2) s per word for μ; ex-Gaussian σ and τ intercepts normal(−2.5, 1) on the softplus scale; random-effect *SD*s Student-*t*(3, 0, *ν*) with $\nu = 0.5$ (Gamma models, ex-Gaussian σ and τ), 1 (logistic and beta-binomial models), or 0.1 (ex-Gaussian μ); random-effect correlations LKJ(2); the Gamma shape gamma(2, 0.5); beta-binomial overdispersion exponential(0.1). Durations were modelled in s per word.

Appendix C

Deviations From the Preregistration

Table C1

Deviations From the Preregistration and Their Effect on the Results

Aspect	Preregistered	As conducted	Reason	Effect on results
Prior-knowledge exclusion	Exclude on prior knowledge (no criterion)	Not applied	No criterion; prequestion scores were low	Minor
Sample size	135 (45 per condition)	138 (47/45/46)	Exclusions left conditions unequal	Minor
DV standardized by length	Not specified	Durations in ms per word	Target sentences differ in length	Minor: pooled comparisons use the same sentences
Regression measure	First-pass regressions	Regressions-in (look-backs)	Error in the preregistration	None: intended measure reported
Inference	$p < .05$ (sidedness not specified)	Bayesian; pd > 95% (97.5% for general effects); Bayes factors	Bayes factors can support the null	Minor: pd > 95% ≈ one-sided $\alpha = .05$
Random effects	Participant intercepts and Topic slopes; some item intercepts	Maximal: plus item intercepts and Status/Group slopes	Maximal structure (Barr et al., 2013)	Minor; conservative
Ex-Gaussian σ	Group × Topic + participant effects	Participant intercepts only	σ not of interest	Minor: σ not interpreted

Aspect	Preregistered	As conducted	Reason	Effect on results
First-pass measure	First-pass duration	Selective regression-path duration	About half of first passes were cut short by a brief return to the preceding line, so first-pass duration was bimodal and no distribution described it	Minor: both measure reading before the reader moves on to later text; selective regression-path duration is not cut short by brief returns
Re-reading model	Linear or ex-Gaussian	Gamma	Re-reading times were close to exponential, so the ex-Gaussian's Gaussian component collapsed	Minor
Final test and prequestion scores	Points per topic, binomial	Points per question, beta-binomial	Allows question random effects	Minor
Specific and general effects	Group × Topic; prequestions vs. control	Status (prequestioned, non-	Separates specific and general effects	None: Group × Topic also reported

Aspect	Preregistered	As conducted	Reason	Effect on results
		prequestioned, control) + Topic		

Note. Effect on results: *minor* = unlikely to change any conclusion. Sample sizes are no prequestions/Polio/TB. None of the preregistered alternatives was analysed, so the effects are judged from the nature of each change rather than estimated.

Appendix D

Drift Correction

Each fixation was assigned to a text line after correcting for vertical drift, separately for each reader and page. The correction combines two approaches used in automated drift-correction algorithms (Carr et al., 2022): assigning sequences of fixations to a line together, and modelling drift as a function of position on the page. First, a single vertical offset was estimated: candidate shifts of the recorded positions were tried in steps of 0.05 line spacings, and the shift that minimized the summed squared distance between each fixation and its nearest text line was kept (distances were capped at one line spacing, so that stray fixations in the margins had little influence). Second, consecutive fixations moving along a line were grouped into chains; a chain ended at a horizontal jump larger than 10% of the screen width (e.g., a return sweep to the next line) or a vertical jump larger than 0.6 line spacings. Each chain was assigned as a whole to the text line nearest its median position, so that a single fixation drifting toward a neighbouring line stayed on the line being read. Third, because drift often grew down the page or across it, a linear model of the recorded vertical position as a function of the assigned line's position and the fixation's horizontal position (recorded $y = a + b \times \text{line } y + c \times x$) was fitted to fixations whose chain lay within half a line spacing of its line; gaze positions were corrected by inverting this model and the chains were reassigned, and this was repeated five times. The linear model was used only if it was plausible (a vertical stretch within 20% and a tilt of less than one line across the screen); otherwise the single offset was kept. Throughout, the correction was limited everywhere on the page to the calibration accuracy criterion of 0.5° (0.57 cm, about 0.45 of a line spacing), so it could not move a reader's gaze by a whole line. Chains more than half a line spacing from every text line were treated as off the text.